

HUM64-3

Sustainable Futures

Planetary boundaries, climate science at decision resolution,
energy-transition economics, climate finance, and the ethics of
intergenerational choice

Reasoning honestly under planetary constraints — neither naïve optimism nor reflexive
catastrophism.

Albert School — 12 hours

Preface

This is the terminal Humanities course, and it does not ask you to start caring about sustainability — it assumes you already do, and asks something harder: that you reason about it with the same discipline you bring to a valuation or a policy memo. The temptation in this domain is to pick a posture. Optimism — technology and markets will solve it — and catastrophism — collapse is imminent and certain — are both ways of **not doing the work**, because each lets a slogan stand in for a number. The standard of this course is the standard of the rest of your degree: distinguish what is measured from what is asserted, carry the uncertainty range rather than the point estimate, and treat sustainability as a **binding input to decisions** rather than as a compliance report filed after the decision is made.

You arrive equipped. From **Macroeconomics** (BUS23-4) you can read the Kaya identity and the difference between relative and absolute decoupling, and you know that a constraint on emissions is a constraint on a system, not a moral exhortation. From **Valuation & Investment Analysis** (BUS43-3) you can build a DCF, estimate a WACC, and — most importantly here — you know that the output of a model is hostage to two or three assumptions, and that the honest deliverable is the **sensitivity**, not the point value. From **International Economics** (BUS43-4) you can do tariff welfare accounting and read a mechanism like the EU’s Carbon Border Adjustment as winners and losers rather than as a headline. From **Labour Economics & Inequalities** (BUS33-4) you have the Gini coefficient, the $r > g$ logic, and — decisively — the causal-inference standards (difference-in- differences, the counterfactual problem, the identifying assumption) that separate a claim from a correlation. From **Business Models** (BUS23-2) you can locate where in a firm’s architecture a cost or a constraint actually bites. And from **Philosophy II** (HUM24-3) you can hold an ethical position under adversarial pressure, name what each framework sacrifices, and recognise — via Jonas — that foresight becomes a moral obligation precisely when consequences are large and irreversible. Climate is the canonical Jonas case.

The course moves in one arc. We begin with the **physical** constraint — the planetary boundaries and the climate science, read at the resolution a decision actually requires, with the honest line drawn between what is well-measured and what is contested. We then move to the **economic** mechanisms that govern the transition — the cost of clean energy and why it surprised everyone, and the corporate and financial machinery (Scope 3, ESG ratings, physical vs. transition risk) through which the constraint reaches a balance sheet. We close with the **ethical** core — the discount rate as a disguised value judgment, and the distributional question of who pays — and then integrate all of it into the S6 quadrangle, where a sustainability constraint becomes an input to a real strategic, financial, and policy decision alongside Strategic Allocation (BUS63-4), Corporate Finance Leadership (BUS63-1), and Public Policy (HUM64-2). By the end the reflex this course exists to build should be automatic: when you meet a green claim, a net-zero pledge, or a doom forecast, never to adopt a posture, but to ask — **what is the number, how wide is the band around it, and what decision does it change?**

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1 Planetary boundaries: what is measured and what is contested

1.1 Nine dials, not one thermostat

Public conversation collapses “the environment” into a single dial labelled **temperature**. The planetary-boundaries framework, proposed by Johan Rockström and colleagues in 2009, refuses that collapse. It identifies **nine** Earth-system processes that together regulate the stability of the planet — climate change is only one — and for each it asks whether humanity is operating inside a **safe operating space** or has pushed the system past a threshold beyond which change may become abrupt or irreversible. The framework’s value is not that it gives nine precise numbers; it is that it forces you to ask, for each boundary, **how confident the number is**. That question — well-measured and defensible, or contested and why — is the entire skill of this chapter.

Definition 1 (planetary boundary) A **planetary boundary** is a proposed quantitative limit on a global environmental variable, set at a level intended to keep the Earth system within the relatively stable conditions of the Holocene. Crossing it does not guarantee disaster, but it marks the edge of the zone in which the system’s behaviour is well understood and reversible. The framework comprises nine boundaries: **climate change**, **biosphere integrity** (biodiversity loss), **biogeochemical flows** (nitrogen and phosphorus), **land-system change**, **freshwater use**, **ocean acidification**, **stratospheric ozone depletion**, **atmospheric aerosol loading**, and **novel entities** (synthetic chemicals, plastics, and other human-made substances).

The honest reading of the framework treats the nine boundaries as differing sharply in **epistemic quality**. Some rest on a well-understood physical control variable with a defensible threshold;

others bundle together processes we can barely measure globally, and their “boundary” is closer to an expert judgment than a measured limit. Figure 1 arranges the nine on exactly these two axes.

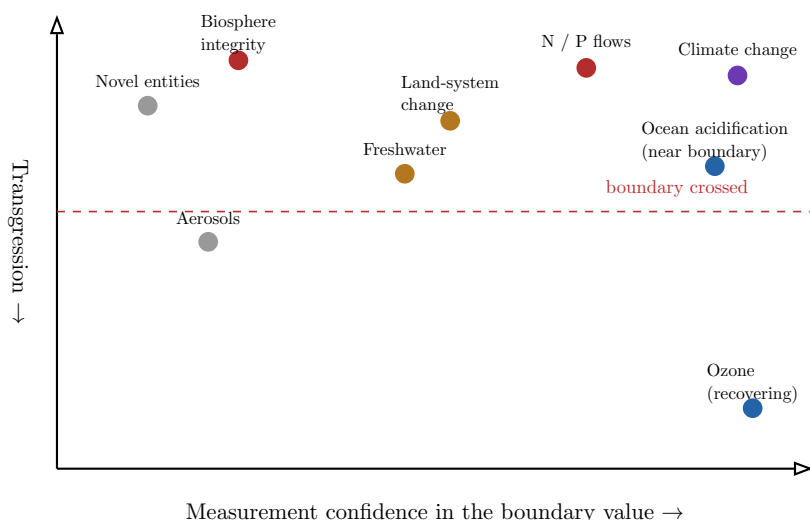


Figure 1: The nine planetary boundaries arranged by **measurement confidence** (horizontal) and **degree of transgression** (vertical). Top-right is the dangerous-and- defensible quadrant: climate change and the nitrogen/phosphorus flows are both clearly transgressed and rest on well-understood control variables. Boundaries on the left — biosphere integrity, aerosols, novel entities — may be transgressed too, but their **thresholds** are contested because the global variable is hard to measure or even define. Confidence in transgression and confidence in the threshold value are different things.

Common pitfall Do not read “boundary crossed” as “catastrophe occurred.” A transgressed boundary signals that we have left the well-characterised safe zone and entered a region of **elevated and poorly bounded risk** — not that a specific disaster is now scheduled. The framework is a risk-management tool, not a doomsday clock. Conversely, do not read “within the boundary” as “fine”: stratospheric ozone is recovering only because of the Montreal Protocol, a policy intervention, not because the problem was never real.

The defensible cases are instructive. **Climate change** has a clear control variable (atmospheric CO₂ concentration, or equivalently cumulative emissions and radiative forcing) tied to a vast, physically grounded evidence base. The **nitrogen and phosphorus flows** boundary is also strongly defensible and **more** transgressed than climate on some accounting: industrial fertiliser has roughly doubled the reactive nitrogen entering the biosphere, with measurable consequences (eutrophication, dead zones). At the other end, **biosphere integrity** is genuinely contested — not because biodiversity loss is in doubt (it is not) but because there is no agreed single global metric and the proposed threshold (extinctions per million species-years) is an expert estimate, not a measured quantity. **Novel entities** — tens of thousands of synthetic chemicals — may be the most under-measured of all: we cannot even enumerate the variable, let alone bound it.

Worked example — Why the contested ones still matter for a decision. Suppose a board must choose between two industrial sites. The climate impact of each can be quantified to a defensible range and priced. The biodiversity and novel-entity impacts cannot be priced with the same confidence — but “we cannot measure it precisely” is not “it is zero.” The disciplined move, following Jonas (HUM24-3), is to treat the poorly-measured-but-plausibly-irreversible impacts under a **precautionary** logic, and to say so explicitly, rather than letting

the measurable crowd the unmeasurable out of the decision entirely. Quantitative honesty includes being honest about what you cannot quantify.

Exercises

1. For each of the nine boundaries, state in one line the control variable that would be measured. Mark the three whose control variable you find hardest to define globally, and explain why that difficulty makes the threshold contested.
2. Stratospheric ozone is the one boundary moving back inside the safe zone. Name the intervention responsible and explain what it tells you about whether boundaries are fixed facts or policy-dependent states.
3. A colleague says “we’ve crossed six of nine boundaries, so collapse is inevitable.” Identify the two distinct errors in this sentence using the distinctions in Figure 1.

Chapter 1 in one paragraph. The planetary-boundaries framework replaces a single temperature dial with nine Earth-system processes, and its real use is to force a separation between **how transgressed** a boundary is and **how well-measured** its threshold is. Climate and nitrogen/phosphorus are transgressed and defensible; biosphere integrity, aerosols, and novel entities are contested in their thresholds, not in the reality of the pressure. Contested does not mean ignorable — it means handled under a precautionary logic, stated explicitly.

2 Climate science at decision resolution

2.1 From “it is warming” to “how much carbon is left”

A general statement — the planet is warming, humans are the cause — is true and, for a decision-maker, almost useless. Decisions need **resolution**: how much warming, for how much more emission, with what range around the estimate, and which consequences are established versus speculative. The IPCC’s Sixth Assessment Report (AR6, 2021–2023) supplies exactly this, and its single most decision-relevant object is the **carbon budget**.

Definition 2 (remaining carbon budget) The **remaining carbon budget** is the cumulative quantity of CO₂ that can still be emitted while keeping global warming below a stated temperature limit with a stated probability. It rests on the robust near-linear relationship between **cumulative** CO₂ emissions and peak warming: to a good approximation, peak temperature depends on the total carbon emitted, not on the path. This is why a budget — a stock — is the right object, and why “we’ll fix it later” is physically expensive: every year of delay subtracts from the same fixed stock.

AR6’s headline numbers, measured from the start of 2020, are worth memorising as orders of magnitude. For a 50% chance of limiting warming to **1.5 ° C**, the remaining budget was about **500 GtCO₂**; for a 67% chance of staying under **2 ° C**, about **1150 GtCO₂**. Global CO₂ emissions run at roughly **40 GtCO₂ per year**. Divide: the 1.5 ° C budget is around a decade of current emissions, and several years of it have already been spent since 2020. The arithmetic, not rhetoric, is what makes 1.5 ° C look out of reach on current trajectories.

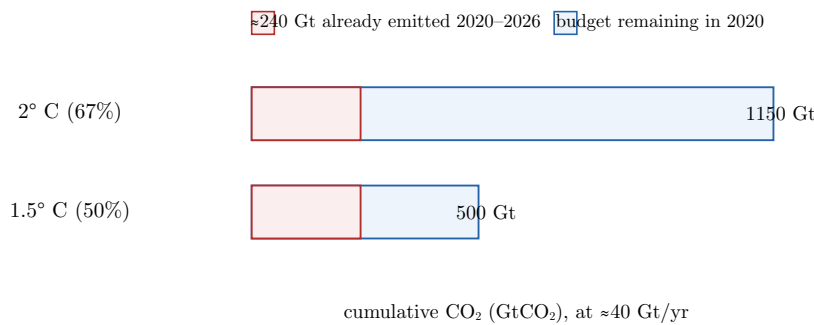


Figure 2: AR6 remaining carbon budgets from the start of 2020. The red segment is the roughly 240 GtCO₂ emitted between 2020 and 2026 at about 40 Gt/yr. Against the 1.5 ° C budget (500 Gt) that segment is already roughly half; against the 2 ° C budget (1150 Gt) it is about a fifth.

Budgets are stocks: the bar does not refill.

2.2 Scenarios and their uncertainty are the message, not a footnote

AR6 frames the future through emissions **scenarios** (the SSP pathways), each producing a **range** of warming rather than a single value. The ranges are not a confession of ignorance to be skipped over — they **are** the decision-relevant content, because they tell you how robust an action is across the futures you cannot rule out. A policy that looks adequate only at the bottom of the range is a bet, not a plan.

Common pitfall The most common abuse of climate science in a business setting is to collapse a range to its midpoint and then treat the midpoint as a forecast. “Warming will be 2.7 ° C” is a category error; “the assessed likely range is roughly 2.1–3.5 ° C under current policies, and our plant siting is robust only below 2.5” is a usable statement. Carry the band. The width of the band is information.

2.3 Tipping points: separating established from speculative

Tipping points — thresholds beyond which part of the climate system shifts to a new state, sometimes irreversibly on human timescales — are where honesty is hardest, because they are simultaneously the most consequential and the most uncertain elements. The disciplined posture distinguishes three tiers.

- **Established and observed:** Arctic **permafrost** is thawing and releasing CO₂ and methane; this is measured, though the rate and feedback strength carry real uncertainty. **Greenland** ice-sheet mass loss is observed and accelerating; its eventual contribution to sea level is large but plays out over centuries.
- **Plausible with weaker constraints:** the **Atlantic Meridional Overturning Circulation** (AMOC) — the current system that includes the Gulf Stream — is assessed by AR6 as very likely to weaken this century, but a **full collapse** this century is judged low- probability and high-uncertainty. Recent papers arguing for an imminent collapse exist but are not the assessed consensus.
- **Speculative:** compound “cascades” in which one tipping point triggers others are physically plausible and worth studying, but quantified probabilities for them do not yet exist at decision resolution.

Worked example — Reading a tipping point like an analyst. An AMOC collapse would be catastrophic for Northern Europe’s climate — but the assessed probability this century is low and the uncertainty wide. The correct decision treatment is **not** “ignore it because it is

unlikely” and **not** “assume it because it is catastrophic.” It is the expected-loss logic you already use: a low-probability, extreme-and-irreversible outcome can dominate a risk calculation even at low probability — which is exactly Jonas’s argument (HUM24-3) for asymmetric caution under irreversibility. State the probability, state that it is poorly constrained, and let the irreversibility, not a point estimate, drive the precaution.

Exercises

4. Using 40 GtCO₂/yr and the AR6 budgets, estimate the year the 1.5 ° C (50%) budget is exhausted on flat emissions, and explain why a declining-emissions path buys more years than the simple division suggests.
5. Classify each of the following as established, plausible-but-weakly-constrained, or speculative, and justify: permafrost methane feedback; AMOC collapse before 2100; a global tipping cascade; Greenland’s multi-century sea-level contribution.
6. Explain why “cumulative emissions determine peak warming” makes the timing of emissions cuts matter less for peak temperature but more for the **budget remaining** to later decision-makers.

Chapter 2 in one paragraph. At decision resolution, the key object is the carbon budget — a fixed stock of CO₂ (≈500 Gt for 1.5 ° C at 50%, ≈1150 Gt for 2 ° C at 67% from 2020) against ≈40 Gt/yr of emissions. Scenarios deliver **ranges**, and the range is the message: carry the band, never the midpoint. Tipping points sort into established (permafrost, Greenland), plausible-but-weakly-constrained (AMOC weakening yes, collapse-by-2100 low and uncertain), and speculative (cascades) — and low-probability irreversible outcomes still earn precaution.

3 Reading a climate-scenario output

3.1 A scenario is an “if”, not a “will”

The IEA’s **Net Zero by 2050 (NZE)** scenario is the document a finance or strategy team is most likely to meet, because it translates the carbon budget into a sector-by-sector pathway: how fast solar and wind scale, when oil and gas demand peaks and falls, how much relies on technologies not yet deployed at scale. The single most important thing to understand about it is grammatical. A scenario is a conditional — **if** these assumptions hold, **then** this pathway delivers the target — not a forecast of what will happen. Your job as a reader is to surface the **structural assumptions** doing the work, find the **uncertainty** around them, and trace the **sector implications** for whatever you are deciding (Learning Outcome 1).

Definition 3 (structural assumption) A **structural assumption** in a scenario is an input that the result is highly sensitive to and that is genuinely uncertain — so that changing it materially changes the pathway. Examples in NZE include the pace of **technology cost declines**, the assumed **deployment rate** of clean capacity, the contribution of **behaviour change** and energy demand reduction, and the scale of **carbon dioxide removal (CDR)** and unproven technologies relied on in later years. Identifying these is the analytical core of scenario reading: it is the same “isolate the two or three assumptions that dominate the output” discipline you learned in DCF sensitivity analysis (BUS43-3).

Common pitfall Two opposite misreadings are equally wrong. The **optimist’s** error is to treat NZE’s feasibility as a prediction that markets will deliver it automatically — it is a normative pathway, not a baseline. The **cynic’s** error is to dismiss it because it relies on some not-yet-scaled technology — every long-horizon scenario does, and the question is not “does it assume anything uncertain?” but “**which** uncertain assumptions is the result most hostage to, and how plausible are they?” Treat scenario reading as assumption-auditing, exactly as you treat a DCF.

Worked example — Three assumptions, three sector implications. Take three NZE structural assumptions and read each as a sector signal for an investor. (1) **Solar/wind cost and deployment continue their steep decline and scale-up** — bullish for renewables, grid, and storage; bearish for thermal-generation equipment. (2) **Oil demand peaks and declines through the 2020s–2030s** — a transition-risk signal for upstream oil assets and reserves that may strand. (3) **CDR scales materially after 2040** — a flag that any pathway leaning on it is making a bet on an immature industry, so a sector exposed to “we’ll capture it later” carries hidden execution risk. Notice the method: each assumption maps to named winners and losers, exactly as a trade-policy episode did in BUS43-4.

The deliverable when you read a scenario is therefore not a summary of its conclusion. It is a short statement of the form: **this pathway delivers X; it is most sensitive to assumptions A, B, C; A and C are the weakest links; therefore for sector S the decision-relevant implication is Y, robust if A holds and fragile if it does not.** That is a usable analytical product; “the IEA says we reach net zero in 2050” is not.

Exercises

7. Pick one NZE assumption you consider a “weak link” and one you consider robust. Justify the classification, and state what evidence would move each.
8. For a named oil-and-gas company, translate the NZE oil-demand pathway into a single transition-risk sentence an investment committee could act on.
9. Explain, using the DCF analogy from BUS43-3, why presenting a scenario’s central pathway without its sensitivity to structural assumptions is the same error as presenting a DCF point value without a sensitivity table.

Chapter 3 in one paragraph. A scenario such as IEA Net Zero by 2050 is a conditional, not a forecast. Read it by isolating the structural assumptions — technology cost and deployment pace, demand/behaviour change, and reliance on immature CDR — that the pathway is most sensitive to, judge which are the weak links, and translate each into named winners and losers for the sector you are deciding about. It is DCF sensitivity discipline applied to a decarbonisation pathway.

4 Energy-transition economics: LCOE and what it misses

4.1 The number that decides a power investment — and the costs it hides

The headline metric of the energy transition is the **Levelised Cost of Electricity (LCOE)**. It answers a clean question — what does a megawatt-hour cost, on average, over the life of a plant

— and its fall for solar and wind is the central economic fact of the last decade. But LCOE is also the most over-quoted number in energy, because it captures the cost of generating electricity **at the plant** and systematically **misses** the cost of turning that electricity into reliable power **on a grid**. Understanding both halves — what LCOE captures and what it omits — is Learning Outcome 2.

Definition 4 (Levelised Cost of Electricity (LCOE)) The LCOE is the constant per-MWh price that, over a plant’s lifetime, exactly recovers all its discounted costs:

$$\text{LCOE} = \frac{\sum_{t=0}^T (I_t + M_t + F_t)/(1+r)^t}{\sum_{t=0}^T E_t/(1+r)^t}$$

where I_t is investment (capital) cost, M_t operations and maintenance, F_t fuel, E_t the electricity generated in year t , and r the discount rate. Two features follow immediately: because the same r discounts costs and output, **capital-intensive**, **zero-fuel** technologies (solar, wind, nuclear) are highly sensitive to r — the cost of capital is a first-order driver of their LCOE — while **fuel-heavy** technologies (gas) are sensitive to fuel prices instead.

Common pitfall LCOE compares technologies **as if a MWh were a MWh regardless of when it is produced**. It is not. A MWh from solar at noon (when many panels produce at once and demand may be low) is worth less than a MWh from a gas plant at 7 p.m. on a still winter evening. LCOE contains **no** information about **when** power arrives or whether it matches demand. Using LCOE alone to declare a system “cheap” is the single most common error in transition economics.

What LCOE misses is, precisely, the cost of **integration**: the extra spending required because solar and wind are **variable** (they produce when the weather allows, not when demand asks) and **non-dispatchable** (you cannot turn them up on command). These costs are real and physical:

- **Intermittency / backup**: when the wind drops, something must fill the gap — storage, flexible gas, or imports — and that backup capacity has a cost the LCOE of the wind farm never counted.
- **Storage**: shifting midday solar to the evening peak requires batteries or other storage, whose cost rises with the duration you need to cover.
- **Grid**: renewables are often built where the resource is (sunny, windy) not where demand is, requiring transmission investment.

4.2 The learning curve that broke the models

For decades, official forecasts repeatedly **over-predicted** the future cost of solar PV, because they treated cost as roughly fixed and modelled the world adding expensive panels. What actually happened is a **learning curve**: each doubling of cumulative installed capacity cut module cost by roughly a fifth (a learning rate of about 20%), and capacity doubled many times. The result is a cost decline of well over 99% since the 1970s — a **structural surprise** to historical models, and the reason a sober reading of the transition must take cost-decline dynamics seriously rather than extrapolating today’s prices flat.

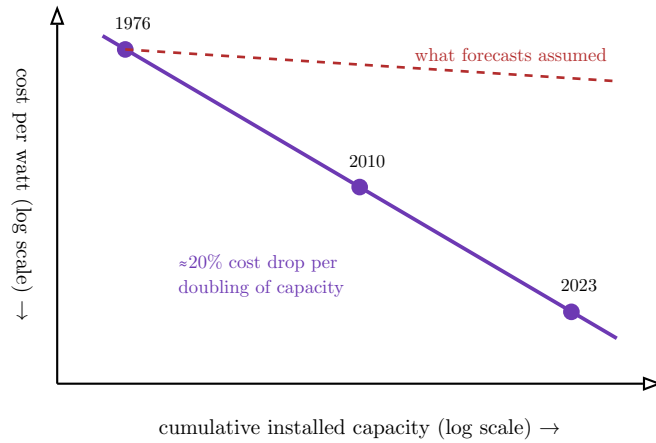


Figure 3: The solar-PV learning curve. On log-log axes the historical cost (solid) falls in a near-straight line — roughly 20% per doubling of cumulative capacity — while official forecasts (dashed) repeatedly assumed cost would stay near the present level. The gap between the two lines is why renewable economics surprised the models: the error was structural, not a one-off miss.

Common pitfall The learning curve is **not** a law of nature and **not** unbounded. It is an empirical regularity that can slow (raw-material limits, interest-rate rises that hit capital-intensive projects, supply-chain shocks) and it applies to the **module**, not the **whole system**. The optimist who extrapolates “costs always fall” into “the transition is basically free” has simply swapped the old error for its mirror image.

4.3 Integration cost: the curve that bends the other way

Here is the physics that LCOE hides and the learning curve does not cancel. As the **share** of variable renewables on a grid rises, the **system** cost of adding the next unit of renewable energy **rises** — gently at low penetration, steeply at high penetration — because more curtailment, storage, backup, and grid are needed to keep the lights on. Generation may get cheaper per panel even as the system gets more expensive per extra percent of renewable share. Both curves are true at once.

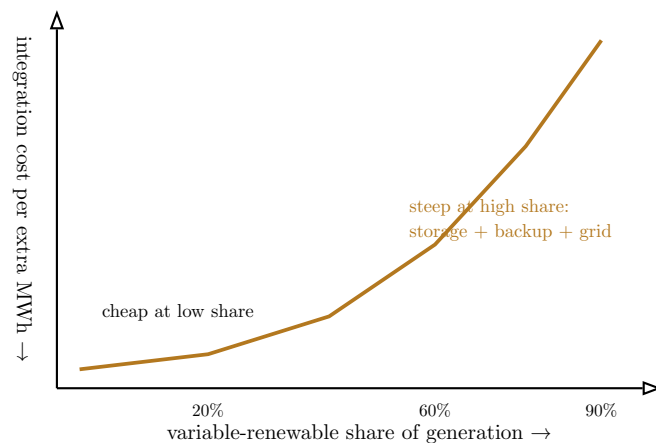


Figure 4: Integration cost rises convexly with variable-renewable share. The first 20–30% of solar and wind integrates cheaply; the last 10–20% is expensive because reliability must be bought through storage, backup capacity, curtailment, and transmission. This is the cost LCOE omits, and it is why “cheapest LCOE” and “cheapest reliable system” diverge at high penetration.

Worked example — Why a falling LCOE and a rising bill can both be real. A utility reports that the LCOE of new solar fell 40% over a decade and is now below new gas. True. Customers also see system charges rise as the grid integrates a high share of renewables. Also true. There is no contradiction: the **plant-level** cost fell (learning curve) while the **system-level** cost of the marginal percent of renewable share rose (integration physics). The analyst’s job is to refuse the headline that uses one number to settle a question that needs both. For investment and allocation over the next decade, the implication is concrete: the cheap-and-easy portion of decarbonised electricity is largely a solved investment case, while value increasingly accrues to the **enablers** of high penetration — storage, grid, flexibility, and dispatchable low-carbon power.

Exercises

10. Write out the LCOE formula and show algebraically why a 2-point rise in the discount rate r raises the LCOE of solar more than that of gas. Tie this back to WACC (BUS43-3).
11. List three costs LCOE omits and, for each, name the technology or sector that profits from supplying the fix.
12. A minister cites the learning curve to argue the transition will pay for itself. Identify which of the two curves in this chapter the argument relies on and which it ignores, and state the honest combined conclusion.

Chapter 4 in one paragraph. LCOE is the discounted average cost of a MWh at the plant; it captures capital, O&M, and fuel, and makes capital-intensive renewables sensitive to the discount rate — but it omits **when** power arrives and the **integration** costs (intermittency, storage, grid) of variable renewables. Two curves govern the transition: the solar learning curve ($\approx 20\%$ cost fall per doubling of capacity, which broke the forecasts) bends costs down, while integration cost bends up convexly with renewable share. A sober investment view holds both: cheap clean generation is largely solved; value migrates to the enablers of high penetration.

5 Corporate sustainability strategy: net zero and the scopes

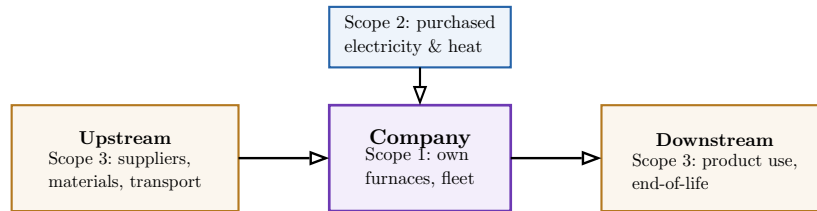
5.1 A pledge is a constraint only if something binds

When a company announces “net zero by 2040,” the analyst’s first question is not whether the target is ambitious but whether it is **binding** — whether it constrains decisions and spending now, or is a marketing signal with no mechanism behind it. This is the same distinction BUS23-2 taught between a business-model cell that is **validated** and one that is merely **assumed**: a net-zero claim is assumed until a dated, costed, accountable mechanism moves it to validated.

Definition 5 (emissions scopes) The GHG Protocol partitions a firm’s emissions into three scopes:

- **Scope 1** — direct emissions from sources the firm owns or controls (its boilers, furnaces, vehicles).
- **Scope 2** — indirect emissions from the **purchased energy** (electricity, heat) the firm consumes.

- **Scope 3** — all other indirect emissions across the **value chain**: upstream (purchased goods, services, transport) and downstream (use of sold products, end-of-life). For most firms Scope 3 is the large majority — frequently **70–90%** — of the total footprint.



Scope 3 (upstream + downstream) $\approx$ 70–90% of the total for most firms

Figure 5: The three emissions scopes. Scope 1 is the firm’s own combustion; Scope 2 its purchased energy; Scope 3 everything else in the value chain, split upstream (suppliers) and downstream (use of sold products). For most firms the value-chain Scope 3 dwarfs Scopes 1 and 2 — which is exactly why it is both the central challenge and the easiest to leave out of a pledge.

5.2 Why Scope 3 is the hard, central problem

Scope 3 is where credibility is won or lost, for three reasons. First, it is **the largest share** of the footprint for most companies, so a target excluding it can omit the bulk of the problem while sounding comprehensive. Second, it is **outside the firm’s direct control** — it lives in suppliers’ factories and customers’ usage — so reducing it requires influencing other actors, not just retooling one’s own plant. Third, it is **hard to measure**: much of it is estimated from spend-based factors rather than metered, with wide error bars and a real risk of **double-counting** across firms (one company’s Scope 3 is another’s Scope 1).

Common pitfall When evaluating a net-zero target, the first thing to check is whether it **includes Scope 3**. A target on Scopes 1 and 2 only — common, because they are controllable and measurable — can leave 80% of the real footprint untouched while the headline says “net zero.” This is not necessarily bad faith; Scope 3 is genuinely hard. But a target that quietly excludes it is, at minimum, not the target a careless reader thinks it is.

Worked example — An oil major and a bank, same word, different meaning. For an oil major, the overwhelming majority of emissions is Scope 3 **downstream** — the CO₂ released when customers burn the sold fuel. A net-zero pledge covering only Scopes 1 and 2 (refinery operations, purchased power) addresses a small fraction of the truth. For a bank, the material footprint is Scope 3 **financed emissions** — the emissions of the companies it lends to and invests in — which dwarf its office energy use by orders of magnitude. In both cases the same phrase, “net zero,” means almost nothing until you ask which scopes it covers. Always force the scope question before the ambition question.

Exercises

13. For three named companies in different sectors (e.g. a carmaker, a software firm, a food retailer), state which scope you expect to dominate and why.
14. Explain the double-counting problem in Scope 3 across a supply chain, and why it makes aggregating Scope 3 across many firms invalid.

15. A firm announces “net zero by 2035.” List, in order, the five questions you would ask to decide whether the pledge is a binding constraint or a marketing signal.

Chapter 5 in one paragraph. A net-zero pledge is a binding constraint only if a dated, costed mechanism backs it; otherwise it is a signal. The GHG Protocol’s three scopes — direct (1), purchased energy (2), value chain (3) — locate where the emissions are, and for most firms Scope 3 is 70–90% of the total. Scope 3 is the central and hardest problem because it is the largest, the least controllable, and the least measurable (with double-counting risk). The first test of any pledge is whether it includes Scope 3.

6 Climate finance: risk, ratings, and impact

6.1 Two ways climate reaches a balance sheet

Climate affects asset values through two distinct channels, and conflating them is a classic analytical error. **Physical risk** is the direct damage of a changing climate — floods, heat, storms, water stress — to assets and operations. **Transition risk** is the risk that the **response** to climate (policy, carbon prices, technology shifts, changing demand) destroys the value of carbon-intensive assets. They can point in opposite directions for the same firm and must be integrated into valuation separately (Learning Outcome 3).

Definition 6 (physical vs. transition risk) **Physical risk:** financial loss from the physical impacts of climate change, split into **acute** (discrete events: a flood, a wildfire) and **chronic** (gradual shifts: sea-level rise, chronic heat, water scarcity). **Transition risk:** financial loss from the move to a low-carbon economy — policy and legal (carbon pricing, mandates), technology (cheaper substitutes), market (demand shifts), and reputation. A **stranded asset** is the paradigm transition-risk outcome: an asset (a coal plant, an oil reserve) that loses value before the end of its economic life because the transition renders it uneconomic.

Worked example — Integrating both into a DCF. Value a coastal refinery using your BUS43-3 DCF. **Transition risk** enters the cash flows: model declining product demand and a rising carbon price as a **bear scenario**, and ask whether the reserve is stranded — i.e. whether terminal value should be written toward zero. **Physical risk** enters separately: a higher probability of flood-driven outages raises operating risk and the discount rate, and may demand capex for resilience. The honest output is not a single number but a **scenario fan** (bull/base/bear) in which the two risks are modelled explicitly — exactly the sensitivity discipline of valuation, now with climate as the dominant assumption. Note the two risks can offset: a firm highly exposed to physical risk may be a transition **winner** (e.g. a flood-resilience engineer), and vice versa.

6.2 ESG ratings: useful, and not what most people think

ESG ratings score firms on environmental, social, and governance criteria, and trillions of dollars reference them. They are also widely misunderstood and legitimately criticised. The most striking empirical fact is **divergence**: the major raters disagree with one another far more than credit-rating agencies do. Where two credit agencies’ ratings of the same bond correlate around

0.99, ESG ratings of the same firm across major providers correlate only around **0.5**. The same company can be rated near the top by one agency and mid-pack by another.

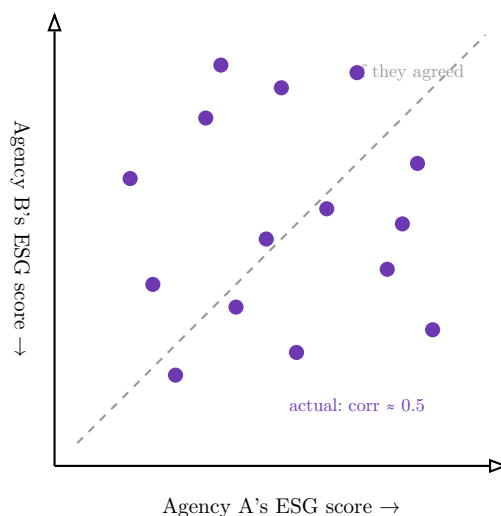


Figure 6: ESG-rating divergence. If two agencies measured the same thing, firms would lie on the dashed diagonal. In reality the scores form a loose cloud (correlation around 0.5, versus 0.99 for credit ratings). The divergence is not mainly noise: it comes from different **scope** (which issues count), different **measurement** (proxies for the same issue), and different **weighting** (how E, S, and G are combined).

The divergence has three identifiable sources, and naming them is Learning Outcome 4. **Scope divergence**: agencies include different issues. **Measurement divergence**: they proxy the same issue differently. **Weight divergence**: they combine the pillars with different weights. On top of this sit three legitimate critiques: **greenwashing** (ratings can reward disclosure and policy language rather than outcomes); the “**S**” **measurement problem** (social factors — labour practices, community impact, human rights — are far harder to quantify than carbon, so “S” scores rest on weaker data); and the **materiality confusion** (some ratings measure ESG **risk to the firm**, others the firm’s **impact on the world** — opposite questions often reported under one label).

Common pitfall A high ESG rating is not evidence of low emissions or positive impact. It may reflect good **disclosure and risk management** relative to sector peers. A tobacco or oil company can hold a respectable ESG score by managing its sector-specific risks well. If you want to know a firm’s climate impact, read its emissions and its capital plan — do not outsource the judgment to an aggregate letter grade whose meaning differs by provider.

6.3 Does ESG investing perform, and does impact investing add anything?

Two empirical questions deserve quantitative honesty rather than ideology. First, **does an ESG strategy outperform?** The evidence is mixed and context-dependent: much measured outperformance is explained by **sector and factor tilts** (ESG funds underweight energy and overweight tech/quality) rather than by ESG information itself, and results are sensitive to period and definition. The defensible statement is that ESG screening does not reliably **sacrifice** returns and does not reliably **generate** alpha; strong claims in either direction usually hide a factor exposure. Second, **does impact investing achieve additionality?**

Definition 7 (additionality) **Additionality** asks whether an investment caused an environmental or social outcome that **would not have happened otherwise**. Buying the green bond of a project that would have been financed anyway provides capital but little additionality; financing a project that could not otherwise raise capital provides real additionality. It is the same **counterfactual** question — what would have happened absent the intervention — that difference-in-differences was built to answer in BUS33-4. Most “impact” claims fail the additionality test precisely because they never state the counterfactual.

Exercises

16. For a named carbon-intensive firm, sketch how physical and transition risk enter a DCF differently — one through cash flows and terminal value, one through the discount rate and capex — and identify which dominates.
17. Two agencies rate a firm A+ and BBB. Using the three sources of divergence, propose the most likely explanation and the one piece of methodology you would read to confirm it.
18. A fund advertises “impact” because it buys green bonds of investment-grade utilities. State the counterfactual that determines whether the investment is additional, and explain why this is a BUS33-4 causal-inference question.

Chapter 6 in one paragraph. Climate reaches a balance sheet through **physical risk** (damage from a changing climate, via cash flows and discount rate) and **transition risk** (response-driven loss, via stranded assets and terminal value); both go into a scenario DCF, and they can offset. ESG ratings **diverge** (0.5 correlation vs 0.99 for credit) because of differing scope, measurement, and weighting, and face real critiques — greenwashing, the hard-to-measure “S”, and the risk-vs-impact confusion. ESG performance claims usually reduce to factor tilts, and impact claims live or die on **additionality** — a counterfactual question, exactly as in causal inference.

7 Net Zero credibility: binding mechanisms vs. decorative offsets

7.1 Carbon neutral, net zero, and the word that does the hiding

The vocabulary of corporate climate claims is engineered for ambiguity, and the analyst’s job is to decode it. The load-bearing distinction is between **reduction** and **offsetting**. A credible target reduces a firm’s own emissions on a steep, dated path; a decorative one leaves emissions roughly flat and buys **offsets** — claimed reductions elsewhere — to declare “neutrality” on paper.

Definition 8 (SBTi validation and the offset question) The **Science Based Targets initiative** (SBTi) validates corporate targets against the emission cuts required to align with 1.5 ° C or well-below-2 ° C pathways. A **binding** target typically has: **absolute** (not merely intensity) reduction goals; **near-term** milestones (e.g. 2030) as well as a distant date; **Scope 3** inclusion where material; and reliance on **actual reductions** rather than offsets for the bulk of the work. A **carbon offset** is a purchased credit representing a tonne of CO₂ reduced or removed elsewhere; offsets are not inherently illegitimate, but their quality varies

enormously and many fail **additionality** and **permanence** tests — the same counterfactual scrutiny as impact investing (Chapter 6).

Common pitfall “Carbon neutral” achieved mainly through offsets is the single most common decorative claim. The red flags: an **intensity** target (emissions per unit of revenue) that lets absolute emissions rise as the firm grows; a target with a far-off date (2050) and **no** near-term milestone, so no current management is accountable; **exclusion of Scope 3**; and heavy reliance on cheap, non-additional, non-permanent offsets (especially unverifiable avoided-deforestation credits). None of these is illegal; all of them should move your assessment from “binding” toward “marketing.”

Worked example — Two pledges, same headline, opposite credibility. Company X: “Net zero by 2050,” intensity-based, Scopes 1–2 only, no interim target, neutrality reached via forestry offsets. Company Y: “SBTi-validated 1.5 ° C target: absolute Scope 1–2–3 cut of 50% by 2030 against a 2020 base, capex reallocated to electrification, offsets used only for residual emissions after 2040.” Both say “net zero.” Only Y has a **binding** structure — dated, absolute, Scope-3-inclusive, costed, with near-term accountability — and only Y’s claim should enter your valuation as a real constraint. Decoding this difference, on a named company, is Learning Outcome 4.

Exercises

19. Build a five-point checklist that separates a binding net-zero target from a decorative one, ordered by how decisive each criterion is.
20. Explain why an **intensity** target can coexist with **rising absolute emissions**, and why absolute targets are therefore the stricter test.
21. Apply the additionality and permanence tests to a forestry-offset-based neutrality claim, and state what evidence would make the offsets credible.

Chapter 7 in one paragraph. A net-zero claim is credible when it is **binding**: absolute (not intensity) cuts, near-term milestones as well as a distant date, Scope 3 included, costed mechanisms, and offsets reserved for residual emissions — ideally SBTi-validated. It is decorative when neutrality is reached mainly by buying offsets of doubtful additionality and permanence while absolute emissions stay flat. The same counterfactual scrutiny used for impact additionality applies to offset quality.

8 Intergenerational reasoning: the discount rate and who pays

8.1 A value judgment disguised as a parameter

Almost every climate-economics conclusion turns on one number — the **discount rate** — and that number is not a technical input but an **ethical choice** wearing a technical disguise. The discount rate sets how much we value harm to future people relative to costs to ourselves now. Choose it high, and a tonne of damage a century out is worth almost nothing today, so aggressive mitigation looks like a bad investment. Choose it low, and that same future damage looms large

today, so aggressive mitigation is urgent. The famous **Stern–Nordhaus** disagreement is, at its core, a disagreement about this single number — and therefore about ethics, not arithmetic (Learning Outcome 5).

Definition 9 (the Ramsey discount rate) The social discount rate is commonly written

$$r = \delta + \eta \times g$$

where δ is the **pure rate of time preference** (how much we discount future welfare simply because it is in the future), g is the per-capita consumption growth rate, and η is the elasticity of marginal utility (how much less an extra euro matters to a richer future person). **Stern** chose a near-zero δ ($\approx 0.1\%$, on the ethical ground that future people’s welfare should not be discounted merely for being future) giving $r \approx 1.4\%$. **Nordhaus** calibrated δ and η to **observed market** returns, giving $r \approx 4.5\%$. Same equation, opposite conclusions — because δ is a value, not a measurement.

The stakes of this choice are enormous and best seen in a single number: the **present value today of €100 of climate damage occurring in 100 years**. Under Stern’s rate it is about €25; under Nordhaus’s it is about €1.20 — a factor of roughly twenty. The entire “act aggressively now” versus “act gradually” divide is largely this factor of twenty.

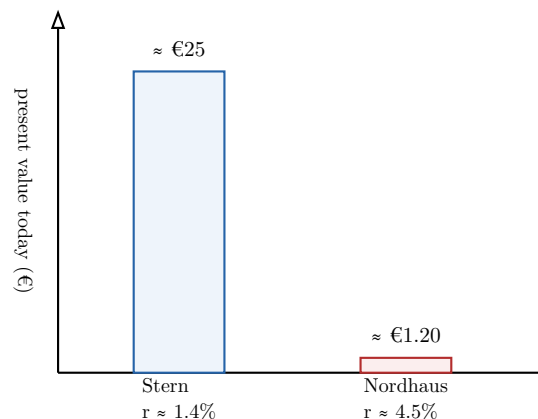


Figure 7: The discount-rate stakes. €100 of climate damage a century from now is worth about €25 today at Stern’s 1.4% but only about €1.20 at Nordhaus’s 4.5% — a factor of roughly twenty. Most of the gap between “decarbonise urgently” and “decarbonise gradually” is not a disagreement about climate science; it is this disagreement about an ethical parameter.

Common pitfall When an economic study reports a “social cost of carbon” or an optimal mitigation path, the **first** thing to find is its discount rate, because it likely drives the headline more than any climate input. A study that buries δ in an appendix and presents its conclusion as objective is hiding the value judgment that produced it. Neither Stern nor Nordhaus is “wrong” — they have made different **ethical** choices, and the honest analyst states the choice rather than smuggling it in as a parameter.

8.2 Who bears the cost: climate justice with quantitative honesty

The second intergenerational and distributional question is **who pays** — across countries and across generations — and here ideology is cheapest and honesty hardest. Three facts can be stated quantitatively. **Historical responsibility is concentrated**: a small set of early-industrialised countries account for the majority of cumulative historical emissions, even though

current annual emissions are now led by large emerging economies — so “who caused it” and “who emits most now” have different answers, and conflating them is a rhetorical move, not an argument. **Vulnerability is inverted**: the countries least responsible historically are often the most physically exposed and least able to adapt. **The transition cost is regressive within countries**: carbon pricing and energy-cost increases fall hardest on lower-income households unless revenue is recycled to offset the burden — which connects directly to the Gini and distributional tools of BUS33-4.

Worked example — Bringing the ethics frameworks to bear (HUM24-3). Climate justice is the canonical case for the ethical frameworks of Philosophy II. A **utilitarian** (Mill) weighs aggregate welfare across all people and generations, which — with a low discount rate — supports strong present action. A **Kantian** asks whether using the atmosphere as a free dump treats future people as mere means. **Jonas** argues that the irreversibility and scale of climate harm make foresight a binding obligation, overriding the comfort of discounting. And — the HUM24-3 discipline — each position must **name its cost**: aggressive mitigation imposes real, regressive burdens on present lower-income people, and an honest climate-justice argument states that burden and addresses it (e.g. revenue recycling) rather than pretending it away. A position that acknowledges no cost is not a defended position; it is a slogan.

Exercises

- 22.** Recompute the present value of €100 of damage in 100 years at $r = 1.4\%$, 3% , and 4.5% , and explain in one line why the gap to the distant future widens so fast with r .
- 23.** Separate “largest cumulative historical emitter” from “largest current annual emitter,” explain why they differ, and state which is relevant to which ethical claim.
- 24.** A carbon tax is criticised as regressive. Using BUS33-4 distributional tools, propose a revenue-recycling design that preserves the price signal while neutralising the regressivity, and state what each ethical framework of HUM24-3 would say about it.

Chapter 8 in one paragraph. The discount rate is an **ethical choice** (the Ramsey rule $r = \delta + \eta g$): Stern’s near-zero δ gives $r \approx 1.4\%$ and urgency; Nordhaus’s market calibration gives $r \approx 4.5\%$ and gradualism — a factor of 20 on the value of distant damage, which is most of the policy disagreement. Distributionally, historical responsibility and current emissions diverge, vulnerability is inverted, and transition costs are regressive within countries unless revenue is recycled. Quantitative honesty and the HUM24-3 rule — every position names its cost — apply throughout.

9 S6 integration: sustainability as a binding input

9.1 The fourth vertex of the quadrangle

This course is the fourth vertex of the S6 integration quadrangle — Strategic Allocation (BUS63-4), Corporate Finance Leadership (BUS63-1), Public Policy (HUM64-2), and Sustainable Futures — and the integration is the point, not an appendix. A sustainability constraint is not a compliance report filed after the strategic decision; it is an **input** to the decision, on the same footing as the cost of capital or the competitive landscape (Learning Outcome 6).

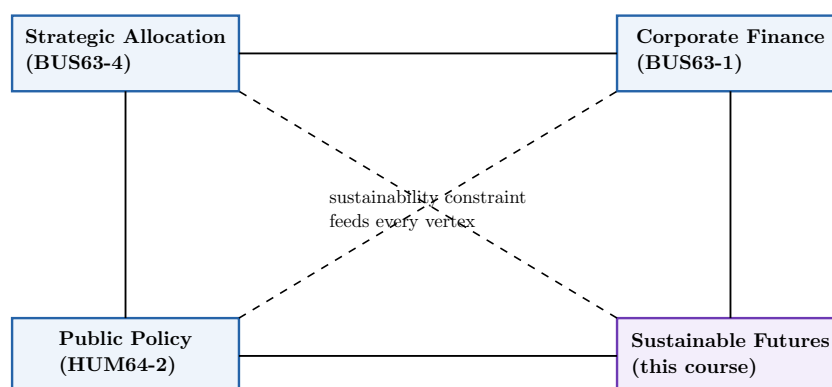


Figure 8: The S6 integration quadrangle. Sustainable Futures supplies a binding constraint — a carbon budget, a transition-risk exposure, a credible-pledge test — that enters Strategic Allocation (which portfolios to fund), Corporate Finance (how to value and finance them), and Public Policy (which instruments to deploy). The constraint is an input to each decision, not a report produced after it.

9.2 Producing and defending a sustainability-constraint analysis

The capstone deliverable integrates everything: read the climate data at decision resolution (Chapters 1–3), apply the economic mechanism (Chapter 4), trace the financial implication (Chapters 5–7), surface the ethical and distributional choice (Chapter 8), and deliver a **defensible recommendation** on a real strategic decision — then hold it under adversarial questioning, exactly as HUM24-3’s jury format trained you to.

Worked example — A worked integration: siting a data centre. A firm must decide where to build an energy-hungry data centre. **Climate data:** read the grid’s decarbonisation pathway and physical-risk exposure (water for cooling, heat) at each candidate site — carrying ranges, not midpoints. **Economic mechanism:** the cheap clean power is real but integration cost and reliability differ by grid; a site on a high-renewable but storage-poor grid carries reliability cost LCOE hides. **Financial implication:** transition risk (future carbon price on any fossil backup) and physical risk (water stress) enter the DCF as scenario cash flows and discount-rate adjustments; a credible corporate net-zero commitment (Scope 2 here being first-order) becomes a **binding** siting constraint, not a press release. **Ethical/distributional:** the local water and grid burden on existing residents is a real cost to name. **Recommendation:** site B, robust across the scenario fan, with a stated dependence on the grid’s storage build-out — and a defence ready for the obvious objection that site A is cheaper on LCOE alone. That last move — anticipating and surviving the objection — is the whole point of the terminal Humanities course.

Common pitfall The failure mode of an integration analysis is to let one discipline silently win: to produce a clean DCF that ignores the carbon constraint, or a passionate climate case with no number, or an ethical stance that names no cost. Integration means **all** the constraints bind at once, the tensions between them are stated, and the recommendation survives an adversary who attacks from each direction in turn. Neither naïve optimism nor reflexive catastrophism survives that room — only quantitative reasoning under acknowledged uncertainty.

Exercises

25. Choose a real strategic decision (a plant, an acquisition, a product line). Produce a one-page sustainability-constraint analysis integrating climate data, an economic mechanism, a financial implication, and the distributional cost — ending in a buy/hold/no recommendation with its key sensitivity stated.
26. For the same decision, write the three sharpest objections an adversary would raise — one attacking your climate numbers, one your financials, one your ethics — and a one-paragraph defence of each that concedes what is fair without abandoning the position.
27. State explicitly how your analysis would feed BUS63-4 (allocation), BUS63-1 (financing), and HUM64-2 (policy) — i.e. what each vertex receives from your constraint.

Chapter 9 in one paragraph. Sustainable Futures is the fourth vertex of the S6 quadrangle, and a sustainability constraint is an **input** to strategic, financial, and policy decisions — not a compliance report. The capstone integrates the whole course: climate data at decision resolution, the energy-economics mechanism, the financial risk channels, the credibility test of any pledge, and the ethical/distributional choice — into a defensible recommendation held under adversarial questioning. The standard, start to finish, is rigorous quantitative reasoning under acknowledged uncertainty: neither naïve optimism nor reflexive catastrophism.